

OSLAB PRACTICAL-9 PROGRAMS & OUTPUT

Date: 03-10-26

Program-1 CODE & OUTPUT:

```
sainath@sainath:~/OSLAB/practical_9$ echo "Hello, this is a file copy test." > source.txt
sainath@sainath:~/OSLAB/practical_9$ ls
copy_low.txt  program1  program1.c  source.txt
sainath@sainath:~/OSLAB/practical_9$ gcc program1.c -o program1
sainath@sainath:~/OSLAB/practical_9$ ./program1
Low-level copy completed.
Time: 0.000057 seconds
sainath@sainath:~/OSLAB/practical_9$ cat program1.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <time.h>

int main() {
    int src, dest;
    char buffer[1024];
    ssize_t n;
    clock_t start, end;

    src = open("source.txt", O_RDONLY);
    dest = open("copy_low.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (src < 0 || dest < 0) {
        perror("open");
        return 1;
    }

    start = clock();

    while ((n = read(src, buffer, sizeof(buffer))) > 0)
        write(dest, buffer, n);

    lseek(src, 0, SEEK_SET);

    end = clock();

    close(src);
    close(dest);

    printf("Low-level copy completed.\n");
    printf("Time: %f seconds\n",
        (double)(end - start) / CLOCKS_PER_SEC);

    return 0;
}
```

```
sainath@sainath:~/OSLAB/practical_9$ ./program1
Low-level copy completed.
Time: 0.000151 seconds
```

Program-2 CODE & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_9$ cat program2.c
#include <stdio.h>
#include <time.h>

int main() {
    FILE *src, *dest;
    char buffer[1024];
    size_t n;
    clock_t start, end;

    src = fopen("source.txt", "rb");
    dest = fopen("copy_lib.txt", "wb");

    if (src == NULL || dest == NULL) {
        perror("fopen");
        return 1;
    }

    start = clock();

    while ((n = fread(buffer, 1, sizeof(buffer), src)) > 0)
        fwrite(buffer, 1, n, dest);

    fseek(src, 0, SEEK_SET);

    end = clock();

    fclose(src);
    fclose(dest);

    printf("Library copy completed.\n");
    printf("Time: %f seconds\n",
        (double)(end - start) / CLOCKS_PER_SEC);

    return 0;
}
sainath@Sainath:~/OSLAB/practical_9$ ./program2
Library copy completed.
Time: 0.000013 seconds
```

Program-3 CODE & OUTPUT:

```
sainath@sainath:~/OSLAB/practical_9$ echo "Hello from input file" > input.txt
sainath@sainath:~/OSLAB/practical_9$ ls
copy_lib.txt  copy_low.txt  input.txt  output.txt  program1  program1.c  program2  program2.c  program3  program3.c  source.txt
sainath@sainath:~/OSLAB/practical_9$ cat program3.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int input, output;
    char buffer[100];

    input = open("input.txt", O_RDONLY);
    output = open("output.txt",
                  O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (input < 0 || output < 0) {
        perror("open");
        return 1;
    }

    dup2(input, STDIN_FILENO);
    dup2(output, STDOUT_FILENO);

    close(input);
    close(output);

    while (fgets(buffer, sizeof(buffer), stdin))
        printf("%s", buffer);

    return 0;
}
sainath@sainath:~/OSLAB/practical_9$ ./program3
sainath@sainath:~/OSLAB/practical_9$ gcc program3.c -o program3
sainath@sainath:~/OSLAB/practical_9$ ./program3
sainath@sainath:~/OSLAB/practical_9$ gcc program3.c -o program3
sainath@sainath:~/OSLAB/practical_9$ ./program3
sainath@sainath:~/OSLAB/practical_9$ cat input.txt
Hello from input file
sainath@sainath:~/OSLAB/practical_9$ cat output.txt
Hello from input file
```

History:

```
547 cd OSLAB
548 ls
549 cd practical_9
550 gcc program1 -o program1
551 gcc program1.c -o program1
552 ./program1
553 gcc program2.c -o program2
554 gedit program2.c
555 gcc program2.c -o program2
556 ./program2
557 cat program2.c
558 ./program2
559 gedit program3.c
560 gcc program3.c -o program3
561 ./program3
562 echo "Hello from input file" > input.txt
563 ls
564 cat program3.c
565 ./program3
566 gcc program3.c -o program3
567 ./program3
568 gcc program3.c -o program3
569 ./program3
570 cat input.txt
571 cat output.txt
572 history
```